Program : **Diploma in Cloud Computing and Big Data**	
Course Code : **5272**	Course Title: **Machine Learning**
Semester : **5**	Credits: 4
Course Category: **Programme core course**	
Periods per week: **4 (L:4 T:0 P:0)**	Periods per semester: **60**

Course Objectives:

- To introduce Artificial Intelligence Concepts
- To impart application of Machine Learning algorithms to make models that predict, group and classify

Course Prerequisites:

Topic	Course Code	Course name	Semester
Calculus and coordinate geometry	1002	Mathematics I	1
Problem Analysis and Programming concepts	2131	Problem Solving and Programming	2
Linear Algebra	2002	Mathematics II	2
Programming in Python	3279	Programming in Python Lab	3
Concept of data, data structures and algorithms	4272	Data Structures and Algorithms	4
Bigdata and data analytics	4271	Big Data Concepts	4
Implementation of data structures	4278	Data Structures and Algorithms Lab	4

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Describe Concept, Types and Challenges of Machine Learning Systems	17	Understanding
CO2	Elaborate Prediction and Clustering Techniques	13	Understanding
CO3	Discuss Classification Techniques	16	Understanding
CO4	Summarize Artificial Neural Network	12	Understanding
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3	1					1
CO3	3	1					1
CO4	3						1

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Describe Concept, Types and Challenges of Machine Learning Systems**		
M1.01	Outline basics of Machine Learning and classify Machine Learning types.	3	Understanding
M1.02	Elaborate Machine Learning Tasks and Performance matrix	6	Understanding
M1.03	Summarize Data Preprocessing techniques	4	Understanding

M1.04	Explain Challenges of Machine Learning	4	Understanding
Contents: **Introduction to Machine Learning, Machine Learning Types-** Supervised, Semi-supervised, Unsupervised and Reinforcement Learning, Online Vs Batch learning, Instance based vs model based. **Machine Learning** - Tasks- Prediction, Classification and Clustering- Applications of each. Training data, Test data, K-fold cross validation. Model parameters, Hyperparameters. Performance Evaluation Metrics - Confusion matrix, Accuracy, Precision, Recall, Specificity, AUC, F-Measure **Prepare the Data for Machine Learning Algorithms-** Data preprocessing- Data Cleaning, Feature Scaling-Standardization, Normalization. Dimensionality reduction-Principal Component Analysis(PCA). **Machine Learning Challenges-** Small dataset, nonrepresentative data, Poor quality of data, Irrelevant Features, Overfitting, Underfitting, Testing and Validation, Hyperparameter Tuning and Model Selection			
CO2	**Elaborate Prediction and Clustering Techniques**		
M2.01	Explain Linear Regression Model for Prediction, and Cost Function- Mean Squared Error (MSE)	2	Understanding
M2.02	Differentiate Gradient Descent, Batch Gradient Descent, Stochastic Gradient Descent, Mini-Batch Gradient Descent.	3	Understanding
M2.03	Compare Regularized Linear Models- Ridge and Lasso and Outline Early Stopping	2	Understanding
M2.04	Classify Detail Clustering - Types, Applications	3	Understanding
M2.05	Describe K-Means clustering model	3	Understanding
	Series Test – I	1	
Contents : Linear Regression for Prediction- Applications, Positive Linear Relationship, Negative Linear Relationship, best fit line, Cost Function – MSE, Residuals, Gradient Descent -batch, mini-batch and stochastic Types of Clustering- Partitioning, Density based, Distribution model based, Hierarchical. Applications of Clustering, K-Means clustering model.			

CO3	**Discuss Classification Techniques**		
M3.01	Describe Logistic Regression	6	Understand
M3.02	Explain Decision Tree and Random Forest	5	Understand
M3.03	Illustrate Support Vector Machine	5	Understand

Contents :

Classification- Binary and Multiclass, Logistic Regression- Training and Cost function, Decision boundaries, Softmax Regression, Cross Entropy, Hyperparameters of Logistic Regression model, Advantages and disadvantages of Logistic Regression.

Decision Tree- Training and Visualizing Decision Tree, Splitting Criteria, Information gain, Gini Index, Stopping Criteria, Pruning methods, Advantages and disadvantages of Decision tree. RDF- ensemble methods, hyperparameters of RDF- advantages and disadvantages of RDF

Support Vector Machine (SVM) – Concept of support vectors, hyper line, and hyperplane, Soft margin and hard margin, Linear and Nonlinear SVM Classification, Outliers, Polynomial- RBF kernel, Similarity Features, Hyperparameters of SVM model, Advantages and disadvantages of SVM classifier.

CO4	**Summarize Artificial Neural Network**		
M4.01	Outline Artificial Neural Network	5	Understand
M4.02	Extrapolate Convolutional Neural Network	5	Understand
M4.03	Summarize Transfer Learning	2	Understand
	Series Test – II	1	

Contents : From Biological to Artificial Neurons, Logical computations with neurons, The Perceptron, Multilayer perceptron(MLP), Backpropagation, Regression and classification MLPs, Fine-tuning Neural network hyperparameters

Deep Learning Concepts, Convolutional Neural Network (CNN), Filters, Convolution and Pooling operation, Vanishing/exploding gradient problems, activation function, Applications of CNN

Transfer Learning Concepts- Introduction to Pretrained CNN architectures- LeNet-5, AlexNet, GoogleNet, VGG, ResNet or Xception

Text / Reference

T/R	Book Title/Author
T1	S. Sridhar, M.Vijayalakshmi, " Machine Learning", Oxford Higher Education
T2	Saikat Dutt, Subramanian Chandramouli, Amit Kumar Das, " Machine Learning", Pearson
R1	Pratap Dangeti, "Statistics for Machine Learning", Packt
R2	Jiawei Han, Micheline Kamber, Jian Pei, “ Data Mining Concepts and Techniques” Third Edition, (MK)Morgan Kaufmann - Elsevier
R3	Josh Hugh Learning, “Python Machine Learning”

Online Resources

Sl.No	Website Link
1	https://cloud.google.com/responsible-ai
2	https://www.geeksforgeeks.org/machine-learning/